

Tightness of contact Godbillon–Vey forms: a classical argument for Calegari’s conditional question

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Abstract

If a global defining form α of a cooriented taut C^2 foliation on a closed three-manifold satisfies $d\alpha = \alpha \wedge \omega$ with a smooth contact form ω , then $\ker \omega$ is tight. The proof combines a special case of Dathe–Khoule’s 2012 general affine-deformation criterion with the Eliashberg–Thurston tightness neighborhood theorem and smooth Gray stability. A finite-interval smoothing argument allows α to be only C^1 . This is a classical corollary, not a claim of a new contact construction or a first historical solution. Priority remains unresolved.

1 The conditional question and the precise claim

The target is Question 13.2 on printed p. 29 of Calegari’s actual 2002 author preprint, dated 8 September, Version 0.78 [1]. Its antecedents are inherited from Question 13.1: $\mathcal{F}$ is a minimal taut C^2 foliation of an atoroidal three-manifold M , and $\text{gv}(\mathcal{F})[M] \neq 0$. One additionally supposes that a choice of global defining form α makes the associated one-form ω contact, with the convention

$$T\mathcal{F} = \ker \alpha, \quad d\alpha = \alpha \wedge \omega. \quad (1)$$

The requested conclusion is that the contact structure $\ker \omega$ is tight. We use the closed, oriented interpretation supplied by a nonzero evaluation on the ordinary fundamental class. This is an interpretation of that evaluation, not a quoted source-wide convention; no boundary or noncompact version is asserted.

Question 13.1 separately asks whether the defining form can be chosen so that $\omega \wedge d\omega$ has one weak sign everywhere. Neither that existence question nor existence of a contact ω is established here. The final AMS chapter published in 2003 has DOI 10.1090/pspum/071/2024640; its full text has not been read, and equality with the 2002 wording is not presumed.

Theorem 1. *Let M be a closed oriented smooth three-manifold, and let $\mathcal{F}$ be a cooriented taut C^2 foliation without spherical leaves. Suppose α is a global nowhere-zero C^1 one-form defining $\mathcal{F}$, and ω is a smooth contact one-form satisfying (1). Then $\ker \omega$ is tight in the usual smooth contact-topological sense.*

Minimality in the source excludes spherical leaves. Atoroidality, minimality, and the numerical Godbillon–Vey value play no further role in this implication. The global defining form supplies the coorientation. The conclusion therefore applies to the specified 2002 conditional question under the smooth contact-form interpretation.

2 Proof by a finite contact path

We use two classical inputs. First, every smooth contact structure sufficiently C^0 close to the tangent planes of a taut C^2 foliation without spherical leaves on a closed three-manifold is tight. This Eliashberg–Thurston neighborhood statement is explicitly recorded in Vogel [4, p. 42] and Dathe–Rukimbira [6, Proposition 3.3, p. 5]; the latter refers to p. 50 of *Confoliations* [3]. We rely on those actual primary-paper statements, rather than claiming a full reading of the book. For negative contact structures apply the same result after reversing the orientation of the relevant component. Second, smooth Gray stability identifies every member of a smooth family of smooth contact structures on a closed manifold by an isotopy [5, Theorem 2.9, p. 2451]. Tightness is consequently constant on such a path.

Proof. Differentiating (1) gives

$$0 = d^2\alpha = d\alpha \wedge \omega - \alpha \wedge d\omega = -\alpha \wedge d\omega. \quad (2)$$

This calculation is valid when α is only C^1 : $d^2\alpha = 0$ distributionally, $\alpha \wedge \omega$ is C^1 , and its product rule yields the continuous form $\alpha \wedge d\omega$. A continuous form that vanishes distributionally vanishes pointwise. For each constant real s , define $\beta_s = \omega + s\alpha$. Then

$$\begin{aligned} \beta_s \wedge d\beta_s &= \omega \wedge d\omega + s(\omega \wedge d\alpha + \alpha \wedge d\omega) + s^2\alpha \wedge d\alpha \\ &= \omega \wedge d\omega. \end{aligned} \quad (3)$$

Thus every β_s is contact, with the same sign. Moreover,

$$\ker \beta_s = \ker(\alpha + s^{-1}\omega) \longrightarrow \ker \alpha \quad \text{uniformly in } C^0 \quad (s \longrightarrow +\infty). \quad (4)$$

Compactness and the positive minimum of $|\alpha|$ justify this uniform plane-field convergence. Choose a finite $S > 0$ for which $\ker \beta_S$ lies in the tightness neighborhood just described.

If α is smooth, (3) gives a smooth contact path on $[0, S]$. Its endpoint is tight, so smooth Gray stability proves the claim at $s = 0$.

For C^1 α , keep this same finite S and keep ω unchanged. Approximate α in C^1 by a smooth one-form a , using smooth charts and a partition of unity, and consider

$$\lambda_s = \omega + sa, \quad 0 \leq s \leq S. \quad (5)$$

Here is a uniform contact bound. Choose a metric and, component by component, an auxiliary orientation making $\omega \wedge d\omega = c(x)dV$ with $c \geq c_0 > 0$. On $M \times [0, S]$, bound both $|\beta_s|$ and $|d\beta_s|$ by B . If the one-form and exterior-derivative errors of (5) are at most δ , their contact-volume error is at most $2B\delta + \delta^2$. Choose δ with this bound below $c_0/2$; the approximation can make the errors at most δ uniformly because S is fixed. Every λ_s is then contact. Also $\ker \lambda_S$ remains inside the C^0 tightness neighborhood by taking the approximation sufficiently small. Smooth Gray stability along (5) transports its tightness to $\ker \lambda_0 = \ker \omega$. $\square$

The approximation a need not define a foliation or satisfy (1). Only closeness on a fixed finite interval is used. Gray stability is never applied to a C^1 path, to $s = \infty$, or to a foliated endpoint. Reversing an auxiliary orientation does not change tightness of the underlying contact planes.

3 Classical attribution and the limits of the note

Dathe–Khoule’s general affine criterion [2, Definition 2.1 and Theorem 2.2, pp. 102–103] already covers the contact mechanism. In dimension three, for an integrable defining form α and a positive contact form ω , its criterion is

$$Q = \alpha \wedge d\omega + \omega \wedge d\alpha \geq 0. \quad (6)$$

Their definition permits $C(t) = 1$, $B(t) = t$, giving $\alpha + t\omega$. In the present case each term of Q is zero, and their expansion specializes to

$$(\alpha + t\omega) \wedge d(\alpha + t\omega) = t^2\omega \wedge d\omega \quad (t > 0).$$

After positive scaling, these are precisely the planes of $\omega + s\alpha$ with $s = 1/t$. The criterion assumes integrability, not closedness of α , and allows zero mixed term; no strictly positive first variation is needed. Negative sign is handled by orientation reversal.

The original 2012 statement was read in an author-uploaded article transcription; its PDF pixels were not obtained. It does not itself print a C^1 threshold. The later coauthor preprint [7, §6, Theorem 6.2, pp. 32–33] explicitly attributes the C^1 criterion to the 2012 work. Its normalized-equality display after equation (62) omits the general nonnegative mixed term and should be an inequality. Equality is correct for $Q = 0$, the only case used here; (3) independently verifies it. No unrelated main result of that 2025 preprint is endorsed.

Theorem 1 is our stated corollary of prior affine, Eliashberg–Thurston, and Gray results. The near-foliation/isotopy tightness argument already occurs with a closed defining form in Dathe–Rukimbira [6, Proposition 3.4]; that narrower statement alone does not cover nonclosed α . No literal statement of the exact tightness answer to Calegari’s Question 13.2 was located in the complete transcribed 2012 article or the relevant 2025 section. This bounded absence is not firstness evidence. The final 2003 chapter, the full *Confoliations* book, the final 2011 Dathe–Rukimbira article, and other identified literature remain unread. Historical priority, novelty of any explicit application or regularity refinement, and present-day open status are unresolved.

The portable support includes the longer current candidate, which separately treats $C^1 \omega$ using an explicitly defined embedded C^2 -disk criterion and tightness of sufficiently C^1 -close smooth contact forms. Those qualified extras are not claims of this short theorem, and are not identified with every low-regularity definition of tightness. Exact symbolic and rational diagnostics check wedge identities, sign conventions, smoothing corrections, and finite-interval margins. They neither prove the imported global contact-topology theorems nor certify priority.

AI tools were used extensively in analysis, drafting, source comparison, and verification. No conventional human peer review or refereeing has occurred. Independent AI audits are not human peer review.

References

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